Natural Language Processing

Natural Language Processing (NLP) is a field of artificial intelligence that enables computers to understand, interpret, and generate human language.

Tokenization divides text into smaller units called tokens. These units may be words, subwords, or characters.

Stemming reduces words to their root form by removing prefixes or suffixes. Lemmatization converts words into their meaningful dictionary base forms.

Applications of NLP include sentiment analysis, machine translation, text summarization, chatbots, and information extraction.

Transformer models use attention mechanisms to understand relationships between words in a sequence.